

Lecture # 10

Digital Communication

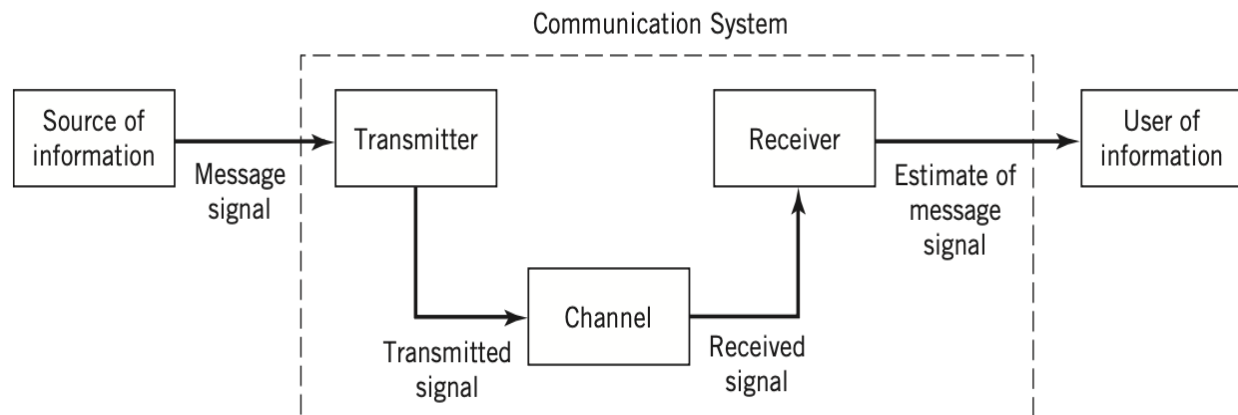
Outline_____

- Basic Digital Communication System,
- A/synchronous Data Transmission, Synchronization
- Probability of Bit Error in Baseband Transmission,
- Matched Filter, and Eye Diagrams

Communication System Basic

❑ Communication System?

- ✓ Communication system is defined as the setup that enables the transmission and reception of information (messages, signals, or data) from one point (transmitter/sender) to another point (receiver) through a communication channel.
- ✓ Irrespective of the form of communication process being considered, there are three basic elements to every communication system, namely, **transmitter**, **channel**, and **receiver**, as depicted in Figure 1.1.



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Figure 1.1 Elements of a communication system.

Communication System Basic

- **Communication System?**

✓ **A good communication system always considers these factors,**

- Collect information from the sender completely
- Transmit this information as much as possible in less time
- Deliver this information to the receiver without any error
- Transmit more information as much as possible.

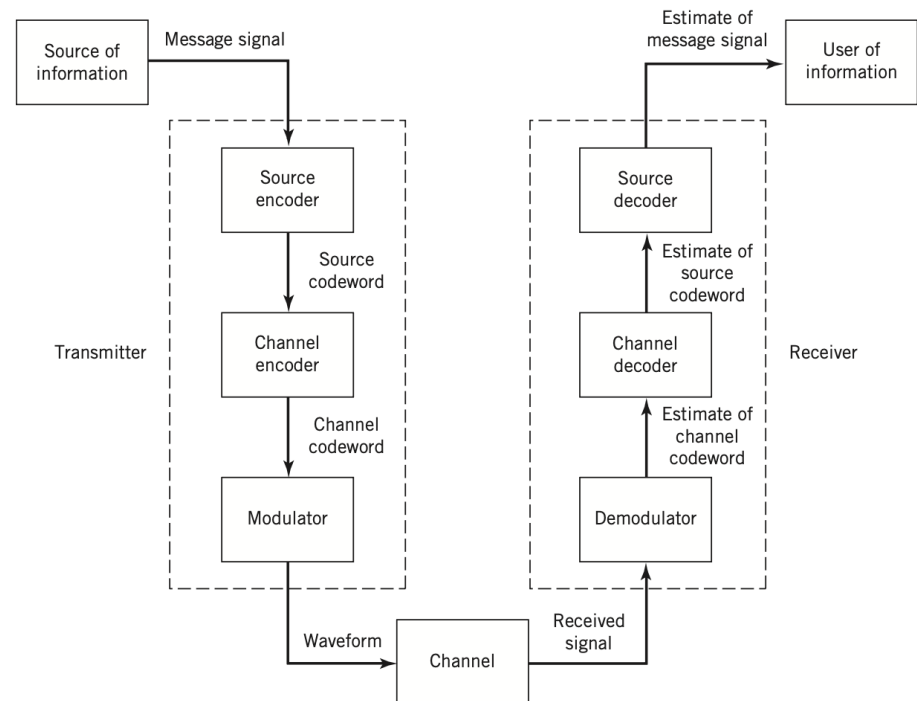


Figure 1.6 Block diagram of a digital communication system.

Transmitter Section

- **Source of Information:** Generates the original analog message signal (like voice or video) or digital data that needs to be transmitted. [PDF](#)
- **Source Encoder:** Compresses the message signal by removing redundant data and converts it into an efficient binary format called a source codeword. [PDF](#)
- **Channel Encoder:** Introduces structured, controlled redundancy bits to the binary sequence to form a channel codeword, which helps detect and correct errors caused by noise. [PDF](#)
- **Modulator:** Maps the digital binary data into distinct physical electrical waveforms or optical pulses suitable for propagating over the physical channel medium. [PDF](#)

The Medium

- **Channel:** The physical transmission medium (such as an optical fiber cable, copper wire, or wireless free space) that carries the waveform but inevitably adds noise and distortion. [PDF + 2](#)

Receiver Section

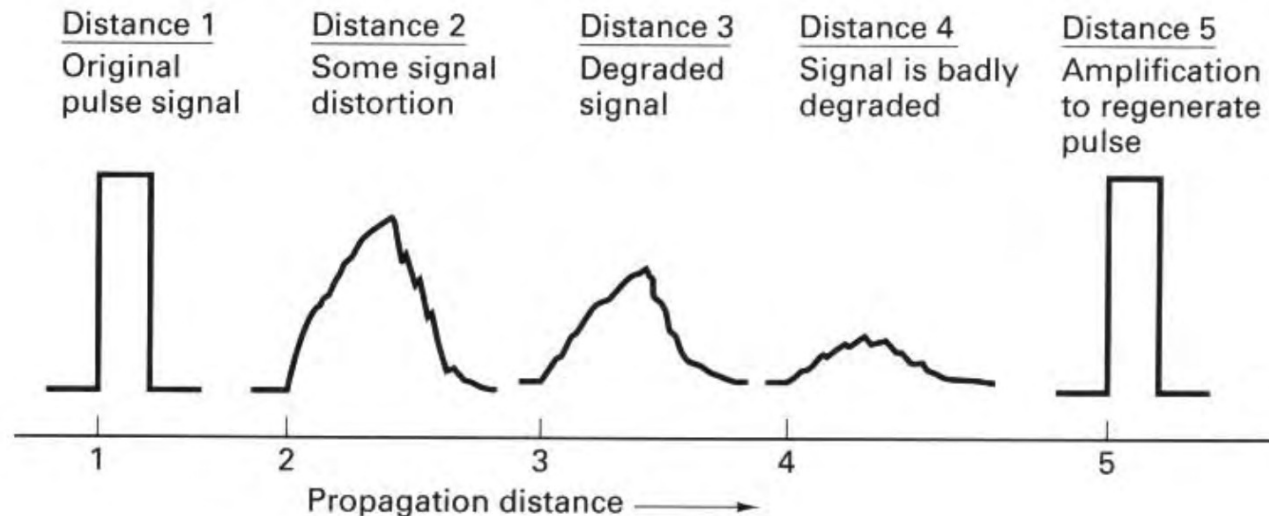
- **Demodulator:** Processes the corrupted received waveform from the channel to extract and recreate the best possible estimate of the binary channel codeword. [PDF](#)
- **Channel Decoder:** Analyzes the added redundant bits in the sequence to detect, isolate, and correct transmission errors introduced by the channel. [PDF](#)
- **Source Decoder:** Reverses the data compression process, converting the cleaned binary bitstream back into its original analog or digital format. [PDF](#)
- **User of Information:** The final destination device or human observer that receives and processes the restored estimate of the original message signal. [PDF](#)

Communication System Basic

□ Why Modern Systems Prefer Digital?

✓ Signal Regeneration

- Digital signals can be **regenerated (reborn)** using repeaters
- Prevents accumulation of noise over long distances



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Figure 1.1 Pulse degradation and regeneration.

Communication System Basic

❑ Why Modern Systems Prefer Digital?

✓ Noise Immunity

- Operates in **two discrete states (0/1)**
- Small disturbances do not affect signal unless threshold is crossed

✓ High Signal Fidelity

- Supports **error detection and correction**
- Enables **very low error rates**, unlike analog

✓ Less Distortion

- Digital signals resist distortion better
- Analog distortion is **permanent and cannot be removed by amplification**

Communication System Basic

□ Why Modern Systems Prefer Digital?

- ✓ **Higher Reliability & Lower Cost**
 - Digital circuits are **more reliable**
 - Easier and cheaper to manufacture (LSI, microprocessors)
- ✓ **Flexibility**
 - Easily implemented with **programmable hardware/software**
 - Supports a wide range of applications
- ✓ **Efficient Multiplexing**
 - **Time-Division Multiplexing (TDM)** is simpler than analog FDM

Communication System Basic

□ Why Modern Systems Prefer Digital?

- ✓ **Uniform Signal Handling**
 - All data types treated equally → *“a bit is a bit”*
- ✓ **Packet-Based Transmission**
 - Data can be sent in **packets**, enabling efficient switching
- ✓ **Security & Processing**
 - Supports **encryption, privacy, and interference protection**
- ✓ **Better for Modern Systems**
 - Ideal for **computer-to-computer communication** and digital devices

Communication System Basic

❑ Costs of Digital Communication System Benefits ?

1. High Signal Processing Requirement: Digital systems are **more signal-processing intensive** than analog systems.

- ✓ They require complex operations like encoding, decoding, compression, and error correction.
- ✓ This increases system complexity, cost, and power consumption.

2. Need for Extensive Synchronization: Digital communication requires **precise synchronization at multiple levels** (e.g., timing, frame, bit level).

- ✓ A large portion of system resources is dedicated to maintaining synchronization.
- ✓ In contrast, analog systems achieve synchronization more easily and with less effort.

Communication System Basic

❑ Costs of Digital Communication System Benefits ?

3. Nongraceful Degradation (Sudden Performance Drop): Digital systems suffer from **nongraceful degradation**:

- ✓ When the **signal-to-noise ratio (SNR)** falls below a certain threshold, performance drops abruptly.
- ✓ The output can suddenly change from **clear and accurate to highly distorted or unusable**.

4. Less Smooth Performance Compared to Analog

- ✓ Analog systems degrade **gradually (gracefully)** as noise increases.
- ✓ Digital systems, however, maintain good quality up to a point, then **fail sharply**, which can be undesirable in some applications.

Communication System Basic

□ Digital versus Analog Performance Criteria:

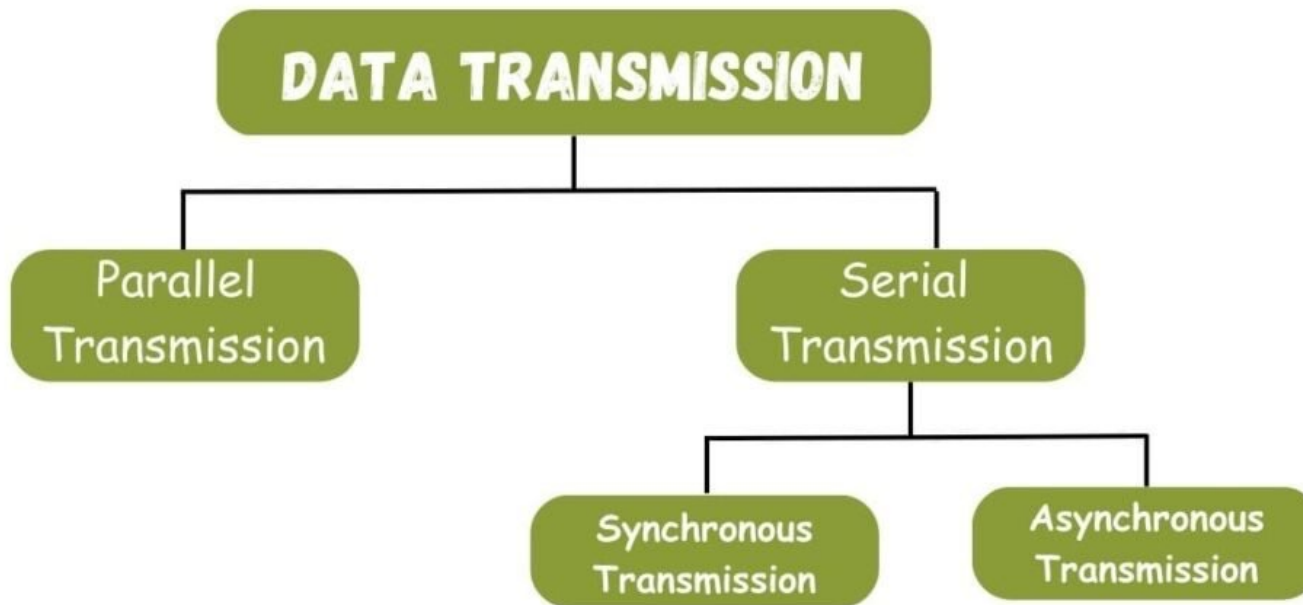
- ✓ A principal difference between analog and digital communication systems has to do with the way in which we evaluate their performance.
 - The figure of merit for the performance of analog communication systems is a fidelity criterion, such as **signal-to-noise ratio, percent distortion, or expected mean-square error** between the transmitted and received waveforms.

By contrast,

- A figure of merit for digital communication systems is the probability of incorrectly detecting a digit, or **the probability of error (P_e)**.

Data Transmission & its Types:

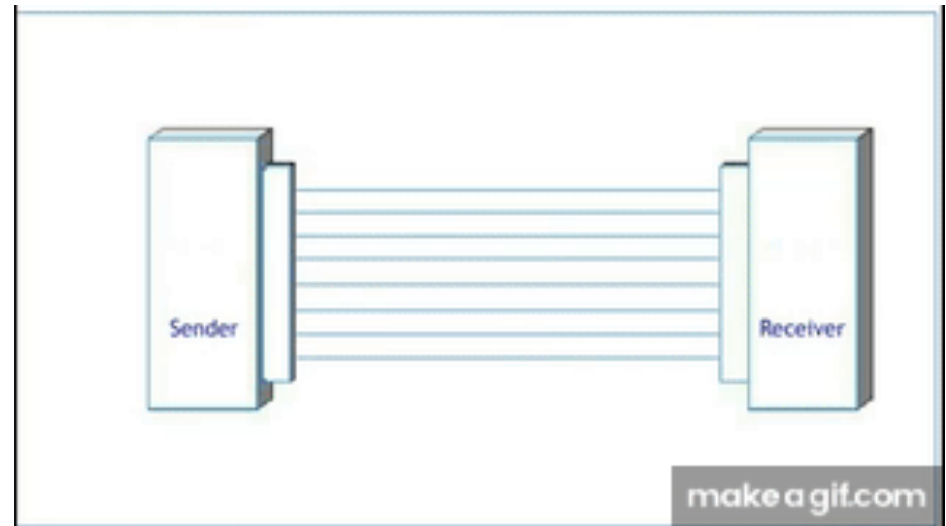
- ❑ **Data Transmission:** It refers to the transfer of data in the form of digital or analog signals from one device to another.



Data Transmission & its Types:

❑ Parallel Transmission:

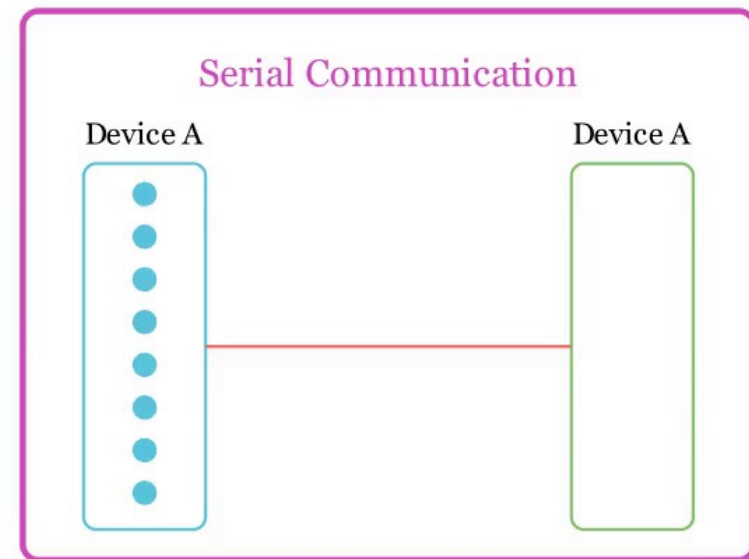
- ✓ This method transmits multiple bits (8, 16, 32, 64 bits) simultaneously across multiple channels.
- ✓ While it allows for faster data transfer rates, it requires **more bandwidth** and is typically used for **short distances**, such as within a computer.
- ✓ Example: VGA port, Printers, etc.



Data Transmission & its Types:

❑ Serial Transmission:

- ✓ In this method, data is sent one bit at a time over a single channel.
- ✓ This approach is simpler and reduces the risk of signal interference, making it suitable for long-distance communication.
- Examples: USB port, Mouse (USB/PS/2), Keyboard (USB/PS/2), Card reader (USB-based)



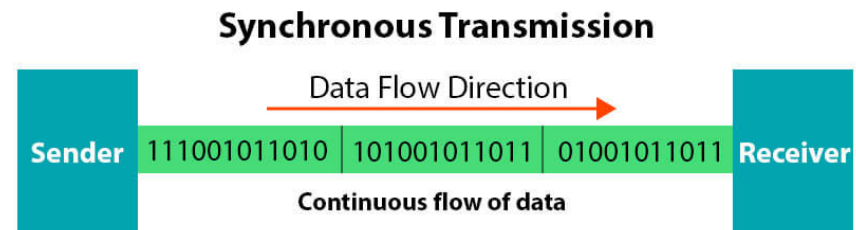
Data Transmission & its Types:

❑ Serial vs Parallel Data Transmission:

Feature	Serial Transmission	Parallel Transmission
Data Transfer Method	1 bit at a time	Multiple bits at once
Speed	Slower per clock cycle, but very high with modern technology	Faster per cycle, limited by distance/interference
Number of Wires	Few (often 1–2)	Many (e.g., 8 or more)
Distance	Better for long distances	Best for short distances
Cost & Complexity	Cheaper and simpler	More expensive and complex
Reliability	More reliable over long distances	Prone to timing errors and crosstalk
Example	Commonly used with the USB (universal Serial Bus) interface	Commonly used with printer cables and other internal device connection

Data Transmission & its Types:

- ❑ **The level of synchronization.** This refers to the degree of syncing between the sender and the receiver.
- ❑ **Data transmission modes:** It can be categorized as either synchronous or asynchronous.
- ✓ **Synchronous (Packet/ Frame/ Block) transmission:** It is a method of data communication in which data is sent as a continuous stream of grouped units—called **packets, frames, or blocks**—that are synchronized between the sender and receiver using a shared timing mechanism (clocking or embedded synchronization signals).



- ✓ constantly,
- ✓ minimizing errors
- ✓ facilitating high-speed, and
- ✓ low-latency communication.

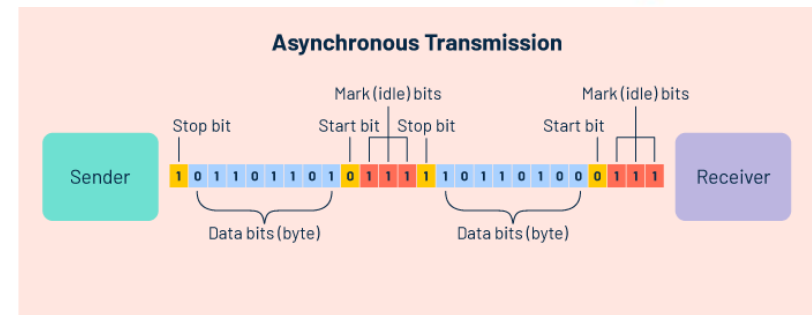
Data Transmission & its Types:

❑ Synchronous (Packet/ Frame/ Block) transmission:

- ✓ **Conceptual structure of a frame:** A typical synchronous frame may include-
 - **Header** (addressing/control information)
 - **Payload/Data**
 - **Trailer** (error detection, e.g., CRC)
- ✓ **Example:** A practical example is **Ethernet communication** using IEEE 802.3 Ethernet:
 - Data from a computer is encapsulated into **Ethernet frames**.
 - Each frame contains:
 - Destination and source MAC addresses (header)
 - Actual data (payload)
 - Error-checking field (CRC in trailer)
 - Frames are transmitted continuously over the network with **synchronization** maintained at the **physical/data link layer**.

Data Transmission & its Types:

❑ **Asynchronous data transmission** is a method of data communication in which data is sent **one character (byte) at a time**, without requiring a shared clock between the sender and receiver. Instead, each data unit is **self-synchronized** using special bits.



✓ **Structure of an asynchronous frame (per character)**

- Start bit (1 bit)
- Data bits (usually 7 or 8 bits)
- Optional parity bit (error checking)
- Stop bit(s) (1 or 2 bits)

Data Transmission & its Types:

❑ Example

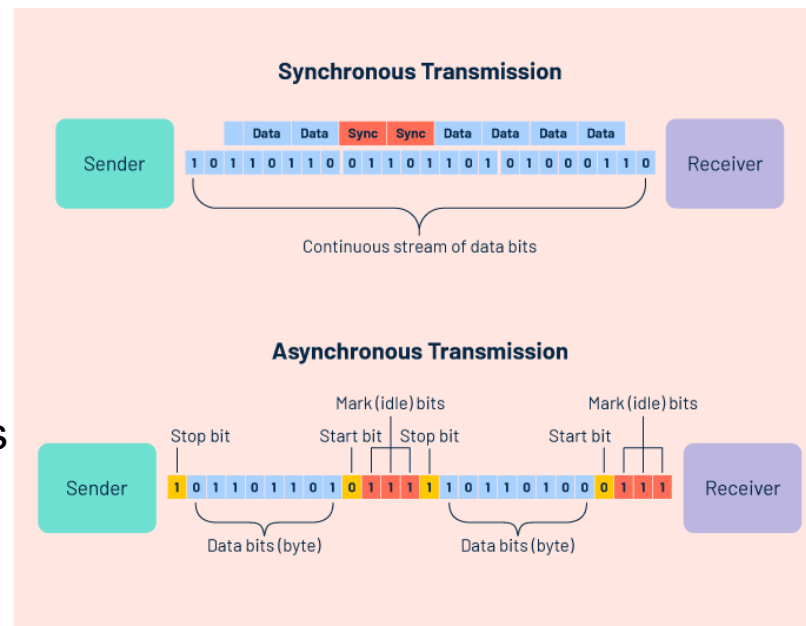
- ✓ A common example is serial communication using RS-232 (Recommended Standard 232):
 - When you type on a keyboard, each character (e.g., “A”) is transmitted individually.
 - For instance, the character “A” (ASCII: 65) is sent as:
 - Start bit → Data bits → Optional parity → Stop bit
 - The receiver detects the start bit, reads the data bits, and uses the stop bit to mark the end before processing the next character.

Data Transmission & its Types:

❑ Real-world example of Synchronization and Asynchronous Example:

Synchronization:

- ✓ USB Flash Drives
- ✓ DDR Memory
- ✓ Serial ATA (SATA) Drives
- ✓ Fiber Channel Storage



Asynchronous

- ✓ Email
- ✓ File transfer
- ✓ Instant messaging
- ✓ Keyboard input

Data Transmission & its Types:

❑ Bit Error Rate (BER):

- ✓ Bit Error Rate (BER) is an important concept to understand in any digital transmission system since it is a major indicator of the health of the system.
- ✓ As data is transmitted some of the bits may not be received correctly. The more bits that are incorrect, the more the signal will be affected.
- ✓ It's important to know what portion of the bits are in error so you can determine how much margin the system has before failure.

Data Transmission & its Types:

❑ Bit Error Rate (BER):

- ✓ BER is the number of bit errors divided by the total number of bits transmitted during a given time interval.

$$BER = \frac{\text{Number of Error Bits}}{\text{Total number of Transmitted Bits}}$$

- ✓ It measures the accuracy and performance of a communication system.
- ✓ A lower BER means better quality and more reliable communication.
- ✓ ***If 1000 bits are transmitted and 100 bits are received incorrectly, what will be BER?***
- ✓ For satisfactory performance, the BER has to be less than 10^{-9}

Data Transmission & its Types:

❑ Factors affecting Bit Error Rate (BER):

1. Interference

- ✓ Increases BER by introducing unwanted signals that distort the transmitted data.
- ✓ Common sources include electromagnetic interference, crosstalk, and multipath fading

2. Increase Transmitted Power

- ✓ Decreases BER by improving the signal-to-noise ratio (SNR).
- ✓ A stronger signal is less likely to be corrupted by noise.

3. Lower Order Modulation

- ✓ Decreases BER because simpler modulation schemes (like BPSK or QPSK) are more robust against noise.
- ✓ Higher-order schemes (like 64-QAM) carry more bits per symbol but are more error-prone.

4. Reduce Bandwidth

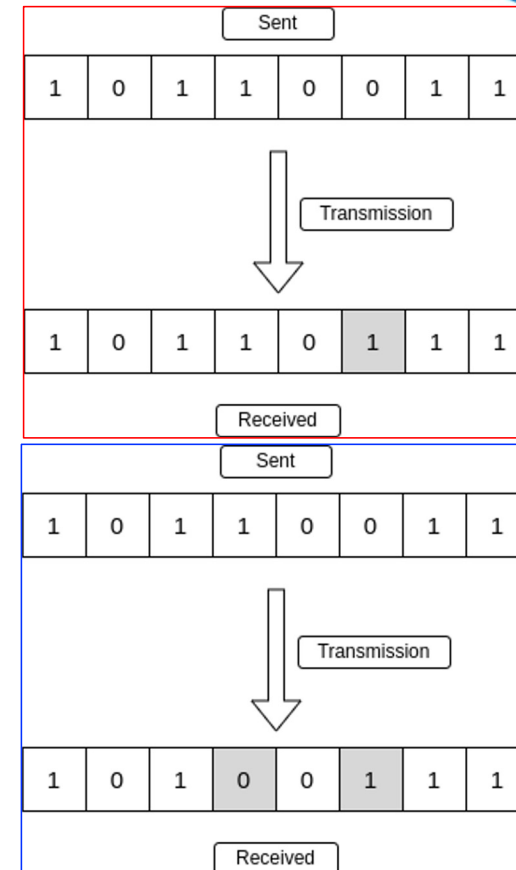
- ✓ Can increase BER if it leads to signal distortion or **inter-symbol interference**.
- ✓ However, in some cases, reducing bandwidth may help filter out noise, depending on system design.

Data Transmission & its Types:

❑ Types of Bit Error?

1. Single-Bit Error: A single-bit error refers to a type of data transmission error that occurs when one bit (i.e., a single binary digit) of a transmitted data unit is altered during transmission, resulting in an incorrect or corrupted data unit.

2. Multiple (spaced)-Bit Error: A multiple-bit error is an error type that arises when more than one bit in a data transmission is affected. Although multiple-bit errors are relatively rare when compared to single-bit errors, they can still occur, particularly in high-noise or high-interference digital environments.



Data Transmission & its Types:

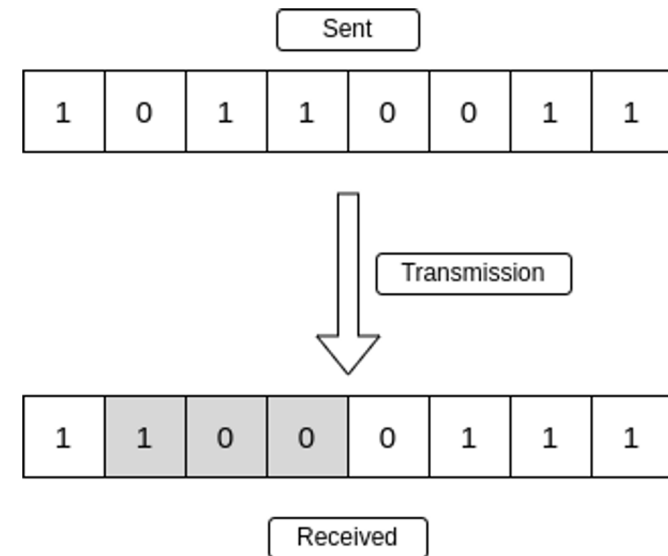
❑ Types of Bit Error?

3. Burst Error

✓ When several consecutive bits are flipped mistakenly in digital transmission, it creates a burst error. This error causes a sequence of consecutive incorrect values.

✓ Burst error are more difficult to detect and correct

❑ **Bit-per-second:** The number of bits of data transmitted every second. It measures data transfer speed (bps, kpbs, Mbps, etc)

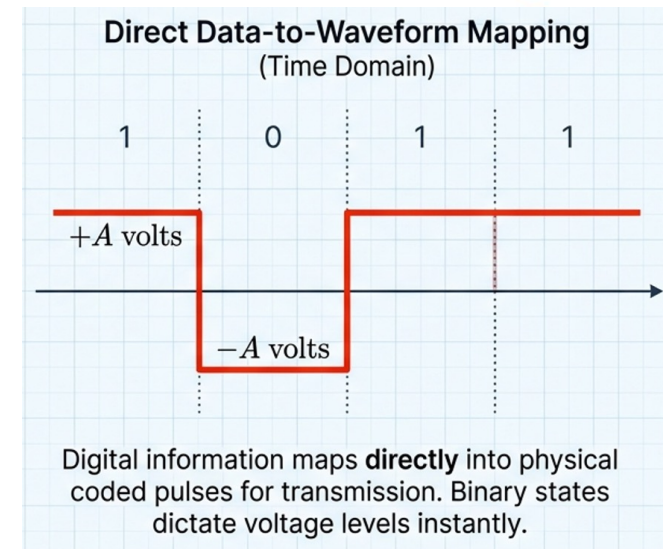


Baseband Transmission:

❑ Probability of Bit Error in Baseband Transmission:

▪ Baseband Transmission?

- ✓ Digital transmission of binary signals directly over a channel without modulation onto a carrier frequency (**Proakis & Salehi**).
- ✓ Transmission of signals in their original low-frequency form, typically using pulse shapes to represent data symbols (**Sklar**).
- ✓ In **baseband digital communication**, information is transmitted using **pulses (waveforms)** that represent binary bits (0 and 1). During transmission, noise—primarily modeled as **additive white Gaussian noise (AWGN)**—distorts the signal.



Baseband Transmission:

❑ Probability of Bit Error in Baseband Transmission:

- The **Probability of Bit Error (BER)** is defined as the probability that a transmitted bit is incorrectly detected at the receiver.

- ✓ Mathematically:

$$P_b = P(\hat{b} \neq b)$$

- ✓ It is one of the most important performance measures of a digital communication system.
- ✓ For binary antipodal signaling (such as polar NRZ, where symbols are represented by +A and -A), the probability of bit error is defined by the **Q-function**

$$P_e = Q \left(\sqrt{\frac{2E_b}{N_0}} \right)$$

Where:

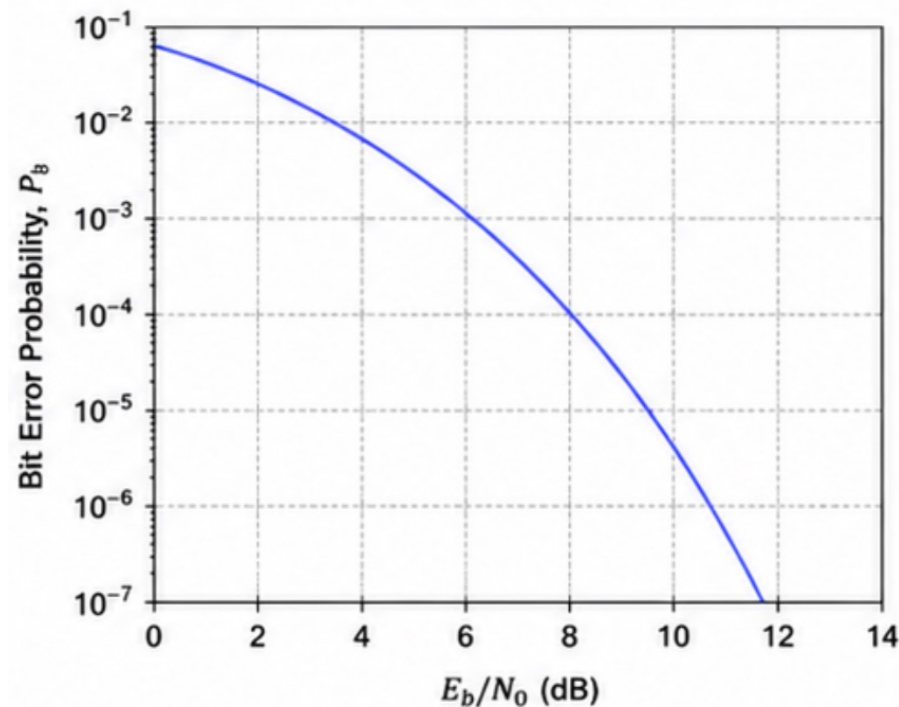
- $Q(\cdot)$ = tail probability of standard Gaussian distribution
- E_b = energy per bit
- $N_0/2$ = noise power spectral density

Baseband Transmission:

❑ Probability of Bit Error in Baseband Transmission:

$$P_e = Q \left(\sqrt{\frac{2E_b}{N_0}} \right)$$

✓ As the E_b/N_0 increases, the area under the Q-function decreases exponentially, leading to a rapid reduction in the bit error rate.



Notice how P_b decreases very rapidly (exponentially) as E_b/N_0 increases.

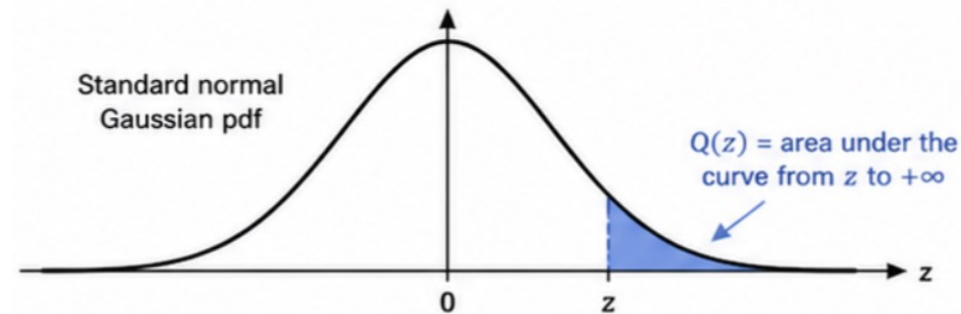
Example

Increasing E_b/N_0 from 4 dB to 10 dB reduces BER from about 10^{-2} to about 10^{-5} ! (1000× improvement)

Baseband Transmission:

❑ Probability of Bit Error in Baseband Transmission:

. The Q-function: area in the tail of a Gaussian distribution



$$Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^{\infty} e^{-t^2/2} dt$$

This is the probability that a standard normal random variable is greater than z .

Some useful values

z	0	1	2	3	4
$Q(z)$	0.5	0.1587	0.0228	0.00135	3.17×10^{-5}

As z increases, $Q(z)$ decreases very fast (exponentially).

Baseband Transmission:

❑ Inter-symbol Interference (ISI):

- ✓ A significant issue associated with baseband transmission is **inter-symbol interference (ISI)**.
- ✓ **ISI** is a form of signal distortion in digital communication where symbols (which are pulses representing one or more bits of data) interfere with adjacent symbols.

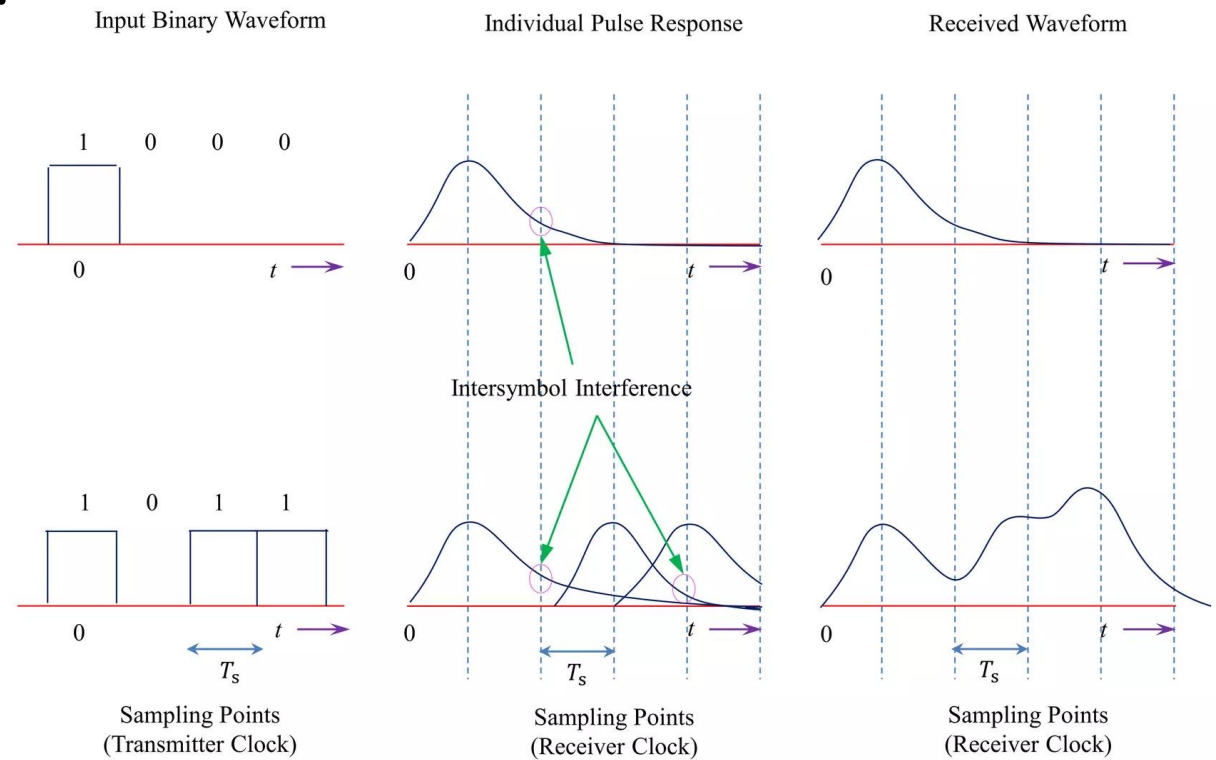


Figure 1: Inter Symbol Interference.

Baseband Transmission:

❑ Inter-symbol Interference (ISI):

- ✓ In an ideal system, each transmitted pulse (representing a symbol, like a '1' or '0') arrives at the receiver distinctly, within its own time slot, without affecting its neighbours.
- ✓ In a real system, due to various factors, these pulses tend to spread out in time.
- ✓ This spreading causes the energy or "tail" of one pulse to spill over into the time slots allocated for neighbouring pulses (either preceding or succeeding ones).
- ✓ This overlap, or interference between adjacent symbols, **is called ISI**.

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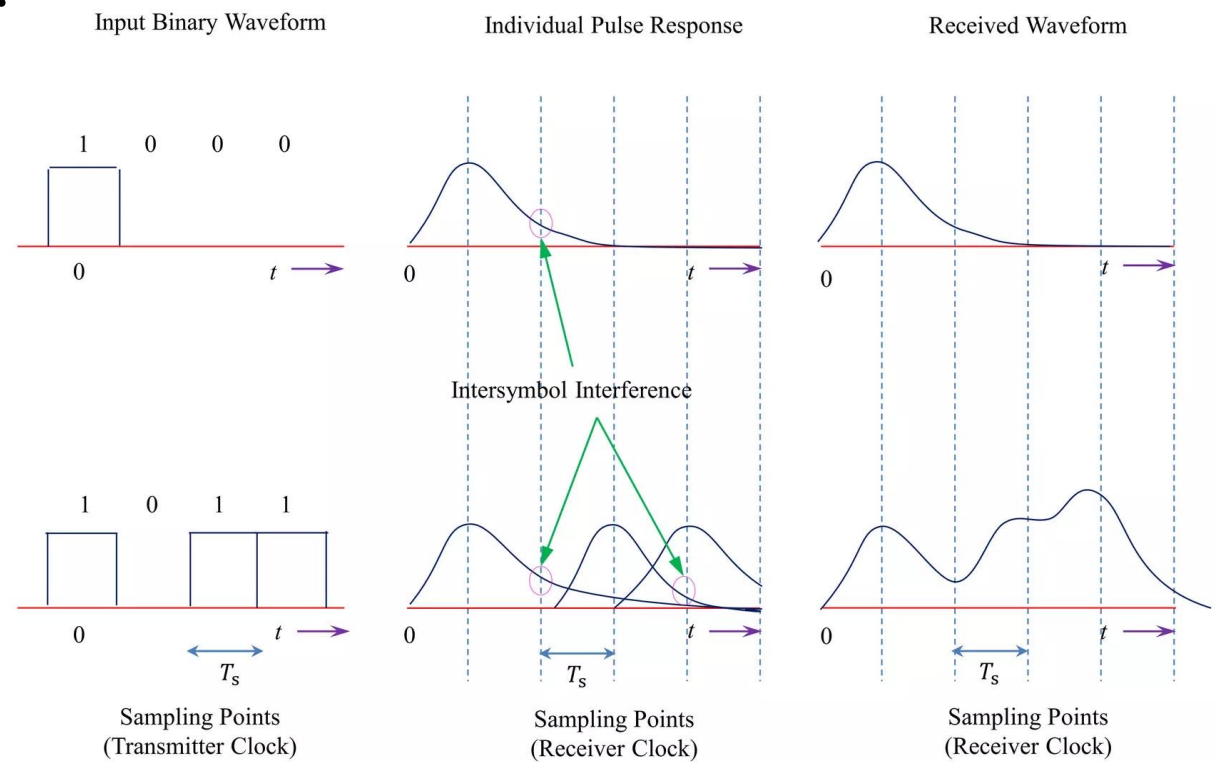


Figure 1: Inter Symbol Interference.

Baseband Transmission:

❑ Inter-symbol Interference (ISI):

- ✓ The consequence of ISI is that it makes it harder for the receiver to correctly detect and distinguish the intended symbol.
- ✓ The received signal for a particular symbol is corrupted by the residual effects of other symbols, increasing the probability of making an error (Bit Error Rate - BER).

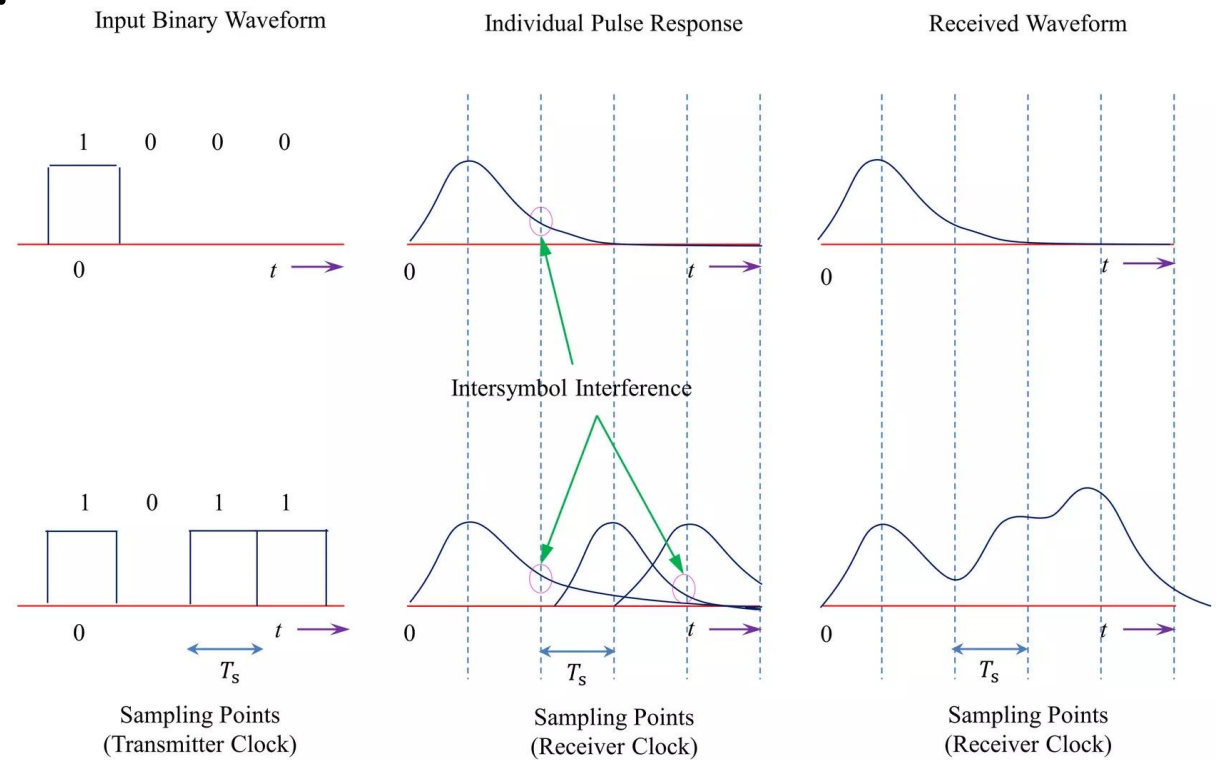


Figure 1: Inter Symbol Interference.

Baseband Transmission:

❑ How is ISI Caused in Communication?

ISI is primarily caused by the characteristics of the physical communication channel and the associated equipment. The main causes include:

✓ **Channel Dispersion** (Bandwidth Limitation and Frequency Response):

- **Limited Bandwidth:** Real-world communication channels (like wires, optical fibers, or wireless medium) inherently have limited bandwidth. They cannot perfectly transmit the sharp, instantaneous changes required for ideal rectangular pulses. Sending pulses through a band-limited channel naturally causes them to spread out in time (become rounded and longer).
- **Non-Ideal Frequency Response:** Channels don't treat all frequencies within their bandwidth equally. Some frequencies might be attenuated more than others, or experience different phase shifts. This distortion in the frequency domain leads to dispersion (spreading) in the time domain.

Baseband Transmission:

❑ How is ISI Caused in Communication?

✓ **Multipath Propagation:**

- This is especially significant in wireless communication but can occur in wired channels too. The transmitted signal travels from the transmitter to the receiver via multiple paths simultaneously (e.g., a direct path plus reflections off buildings, terrain, or other objects).
- These different paths have different lengths, meaning copies of the signal arrive at the receiver at slightly different times (with different delays).
- When these **delayed copies** overlap with the primary signal or subsequent symbols, they cause ISI. The receiver sees a composite signal blurred by these delayed versions.

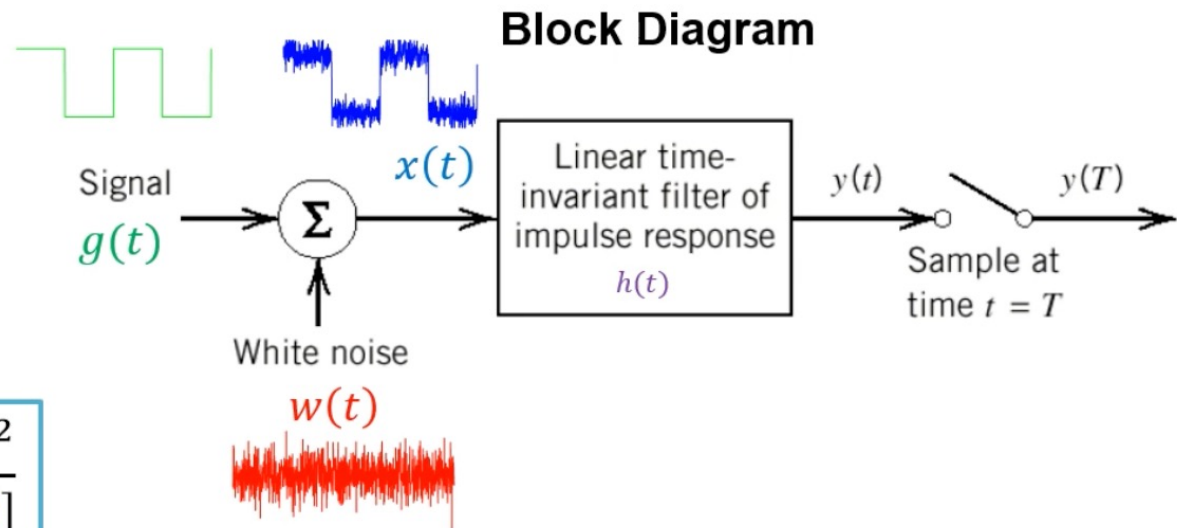
- ✓ **Filtering:** While filters are often used intentionally in transmitters (for pulse shaping to *limit* interference) and receivers (matched filters to *maximize* signal-to-noise ratio), imperfect filter design or the combination of transmit/receive filters with the channel response can inadvertently contribute to or worsen ISI.

Baseband Transmission:

❑ Matched Filter: Control or mitigate ISI at the Receiver!

- ✓ To achieve optimal detection of the signal at the receiver end, a **linear time-invariant (LTI) filter**, known as a **matched filter**, is used.
- ✓ A **matched filter** is a linear time-invariant filter designed to detect a known signal pulse in the presence of noise by maximizing the output **signal-to-noise ratio (SNR)** at a specific sampling instant.

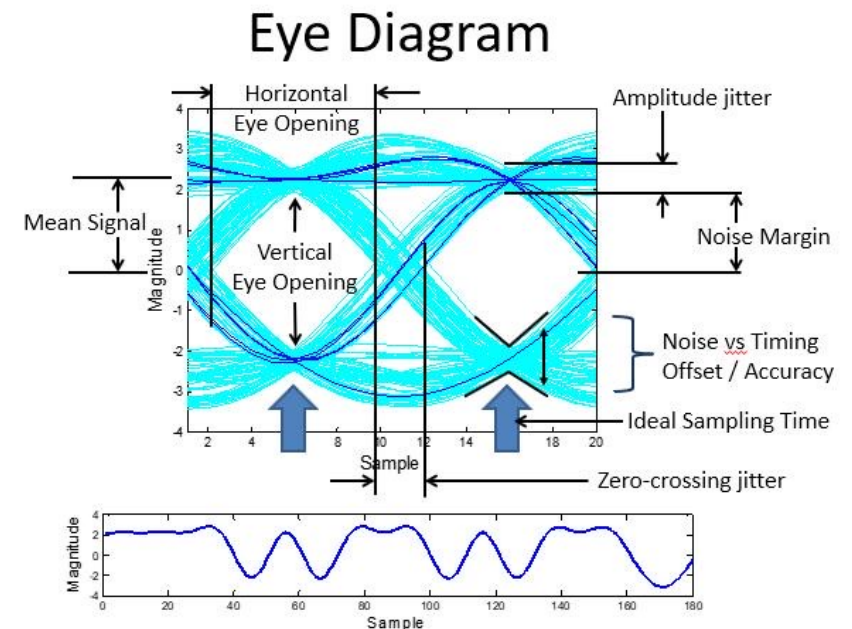
$$\text{Maximize } \eta = \frac{|g_o(T)|^2}{E[n^2(t)]}$$



Baseband Transmission:

□ Eye Diagram:

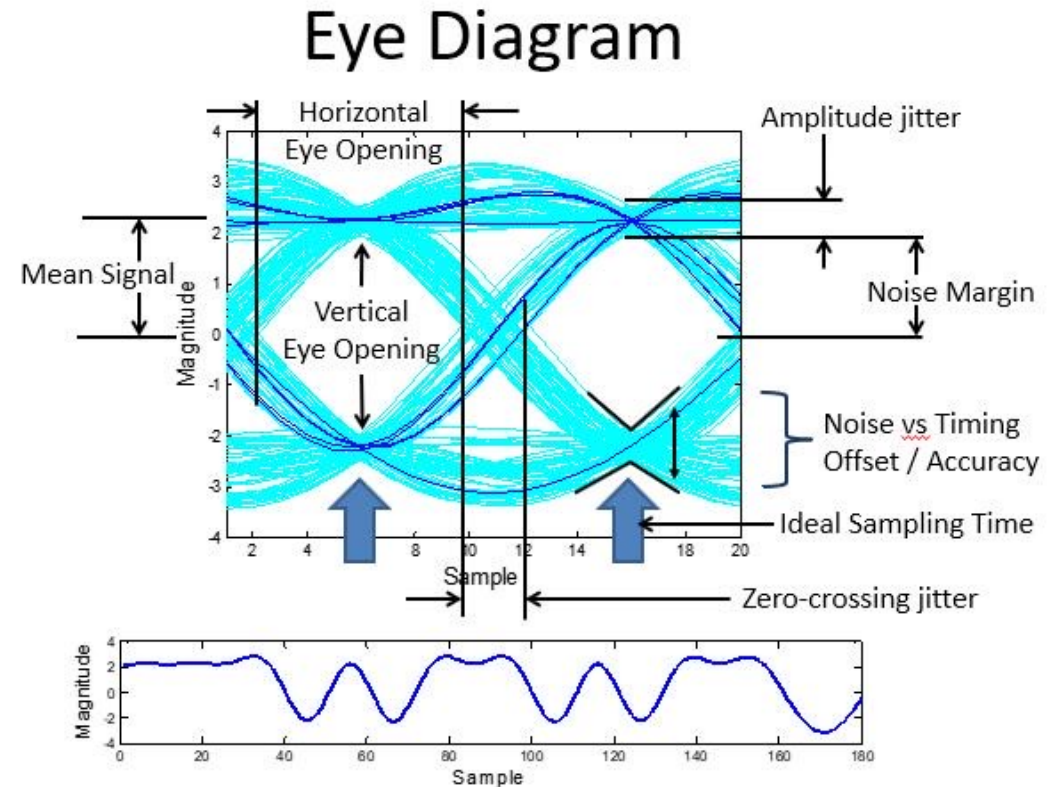
- ✓ A common method of analyzing the **quality of binary data transmitted** on a cable is to display what is known as the **eye diagram** on a common oscilloscope.
- ✓ The **eye diagram, or pattern**, is a display of the individual bits overlapped with one another. The resulting output looks like **an open eye**.
- ✓ The shape of the pattern and the degree that the eye is “open” can be used to determine many things about the quality of transmission.



Baseband Transmission:

□ Eye Diagram:

- ✓ An **eye diagram** is used to **visually evaluate the quality of a digital communication signal**, especially in the presence of **noise, distortion, and inter-symbol interference (ISI)**.
- ✓ An **open "eye"** signifies minimal distortion and reliable data transmission, while a **closing eye** indicates increased Inter-Symbol Interference (ISI) and higher bit error rates (BER)



Baseband Transmission:

□ Eye Diagram:

- ✓ **Formation:** It is formed by synchronizing the oscilloscope to the data clock and overlapping consecutive bit intervals, showing all possible 0 and 1 transitions.
- ✓ **Interpretation:**
 - **Eye Opening:** A wide, open eye indicates low ISI and high signal amplitude, representing a healthy signal.
 - **Eye Closure:** A closed or narrow eye indicates high distortion, noise, and potential timing errors.
 - **Jitter:** The horizontal thickness of the signal crossing points represents timing variations (i.e., jitter).
 - **Noise/Amplitude:** The thickness of the top and bottom lines represents amplitude noise or inter-symbol interference.

